

Ziyun Lan

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Education

Aug 2026 – Present	National University of Singapore	Computing (Computer Science) Master
Sep 2022 – Jun 2026	Nanjing Normal University	Computer Science and Technology Bachelor

• **Cumulative GPA:** 3.9/5, **Ranking:** 5/113

• **Core Courses:** Java Programming, Machine Learning, Algorithm Design, Computer Organization and Architecture, Operating Systems

Project Experience

Jul 2026 – Aug 2026	RAG and Agent-based Intelligent Agent for Robot Vacuum Cleaners	Personal Project
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• **Project Overview:** Developed a RAG-Agent Q&A system with retrieval & tool-calling for Q&A and report generation.

• **Responsibilities:** 1. Built ReAct Agent via LangChain, wrapping RAG, weather, location & usage-record tools; 2. Deployed Chroma for document loading, recursive splitting, embedding & recall; 3. Implemented context tagging & dynamic prompts for personalized reports using external data; 4. Created Streamlit UI with logging, error-handling & MD5 deduplication for incremental KB updates.

Jul 2026 – Aug 2026	EyeVQA: Ophthalmic VQA Benchmark for Recognition and Spatial Grounding	MICCAI Workshop (Oral)
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• **Project Overview:** Constructed an ophthalmic VQA benchmark for disease recognition, cross-image reasoning and spatial grounding.

• **Responsibilities:** 1. Aggregated 21 ophthalmic datasets; preprocessed images in Python, compiled 27,935 fundus images and constructed 20,000 JSON-formatted QA pairs; 2. Generated ground truth from diagnostic labels, severity grades, mWasks and anatomical annotations, designed 7 question types; 3. Conducted zero-shot evaluation of 14 VLMs and computed metrics including accuracy and IoU..

Sep 2024-Jul 2025	YOLO-based Intelligent Parasite-Egg Detection System for Microscopic Images	Nanjing CDC Project
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• **Project Overview:** Developed an automatic pipeline eliminating inefficiency and errors in manual microscopic parasite egg observation.

• **Responsibilities:** 1. Optimized image stitching via SIFT feature matching and slope filtering to build global coordinate system; 2. Trained YOLOv8n model with active learning for fast parasite egg detection; 3. Realized egg deduplication through coordinate transformation and Euclidean distance algorithm; 4. Integrated interfaces to achieve automatic accurate counting, panorama and coordinate output.

Internship Experience

Oct 2025-Mar 2026	The 14th Research Institute of CETC	Software Development / Algorithm Intern
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AFSIM Parsing: Leveraged Qt for recursive file parsing & UI interaction, regex-extracted config to link platforms, routes and sensors.

Aircraft Trajectory Classification System: Built trajectory classification visualization for rule-based prediction limitation. 1. Generated normalized-image trajectory samples with Python; 2. Built PyTorch-based CNN with data augmentation; 3. Designed binary-anomaly + fine-grained multi-class pipeline, visualized outputs with Matplotlib.

Rag-base Q&A System: Deployed Ollama, RAGFlow & local LLM on Docker, implemented chunking, vector indexing & QA interface.

Aug 2025-Oct 2025	Nanjing Citizen Card Payment Co., Ltd.	Robotic Arm Development Intern
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Robotic-Arm Environment Setup: Documented JetArm hardware & wiring, configured Linux and ROS2, deployed NoMachine and SSH.

Citizen-Card Sorting & Counting System: 1. Researched and adapted lightweight YOLO, tuned parameters to improve card-positioning accuracy; 2. Used Python to fetch 3D coordinates from depth camera, optimized height logic for precise robotic-arm positioning; 3. Tuned motion & gripping parameters to realize stable grasping and sorting; 4. Built counting module for automatic total-quantity statistics.

Mar 2025-Aug 2025	Nanjing Borun Brain-Inspired Intelligence Technology Co., Ltd.	Algorithm Intern
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Helmet Object-Detection: Performed Labelme annotation for YOLO, split datasets & configured YAML for auto-training and surveillance.

Multi-Label Behavior Classification Model: 1. Modified YOLO to lift single-label constraints; 2. Customized network with new activation & loss functions for multi-scenario adaption; 3. Built ResNet18 multi-label model with Dropout to mitigate overfitting; 4. Developed API-&-LLM-powered auto-annotation scripts for label refinement; 5. Used Git for version control & Docker for environment isolation.

Honors & Awards

• **Tech Contests:** Provincial 3rd, Computer Design Contest | Twice Provincial 3rd, Lanqiao Cup C++ | Provincial 1st/2nd, National C++ Challenge | Provincial 1st, Advanced Office Contest | National 3rd, Computer Skill Contest | Provincial 2nd, Java Programming Summit

• **Math Contests:** Twice Provincial 2nd, Huajiao Cup Math Contest | Provincial 2nd, National Math Ability Challenge

• **Honors:** 1st/2nd-class Student Scholarships (3 times) | Merit Student | 2 software copyrights

• **Certifications:** Software Designer (Intermediate) | MIIT-certified Internet Platform Dev Engineer | NCRE Level-2 | IT Teacher Qualification

Technical Skills

• **Skills:** **Languages:** Python | C++ | Java | Verilog; **Deep Learning:** TensorFlow | PyTorch | OpenCV | YOLO | CNN | RNN | LSTM | Transformer;

Frontend: HTML | CSS | JavaScript | React | Vue; **Backend & Frameworks:** Spring Boot | MyBatis | Qt; **Databases:** MySQL | PostgreSQL; **Agents:** LangChain | RAG | Agent | Prompt | Codex | Trae; **Tools & Platforms:** Git | Docker | Linux | Notebook | Nginx | PPT | Word | Excel | Overleaf

• **Language Proficiency:** IELTS 6.5 (6 in each sub-score); GRE 322 (AW 3.5); CET-6;